

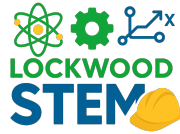
Introduction to Engineering Design

Aerospace Academy Version

Unit 2 Rubric

Introduction to Engineering Design

Course: Introduction to Engineering Design
Program: Academy of Advanced Aerospace Engineering
Brand: Lockwood STEM
Use: Student-facing and teacher grading rubrics



Unit 2

Technical Sketching for Aerospace Design

Concept sketching, isometric sketches, multiview drawings, line conventions, dimensioning, and final aerospace drawing package.

Grading Rubrics and Scoring Guide

Unit 2: Technical Sketching for Aerospace Design

Assessment Purpose

Concept sketching, isometric sketches, multiview drawings, line conventions, dimensioning, and final aerospace drawing package. This rubric is designed for consistent grading, feedback, and student self-checks.

Suggested 4-Point Scale

4 = Expert: exceeds expectations with clear evidence, precision, and independence. **3 = Proficient:** meets expectations with complete and accurate work. **2 = Emerging:** partially meets expectations; work is incomplete, vague, or inconsistent. **1 = Incomplete:** missing, unsafe, inaccurate, or shows limited evidence of understanding.

Unit 2 Score Summary

Category	Points Possible	Points Earned
Sketching setup and craftsmanship	10	
Concept sketching	10	
Isometric sketches	15	
Multiview drawings	15	
Line conventions	10	
Dimensioning	15	
Annotations and design notes	10	
Aerospace drawing package	15	



Lockwood STEM

Academy of Advanced Aerospace Engineering

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Unit 2 Project Rubric

Criterion	Pts	4 – Expert	3 – Proficient	2 – Emerging	1 – Incomplete
Sketching Setup & Craftsmanship	10	Work is neat, dark enough to read, aligned to graph paper, and uses appropriate tools/line quality.	Work is readable and mostly neat.	Work is messy, faint, or inconsistently aligned.	Work is missing or unreadable.
Concept Sketching	10	Concept sketches communicate design intent with labels, annotations, and multiple viewpoints when useful.	Sketches communicate the main idea with some labels.	Sketches are incomplete or difficult to interpret.	Missing.
Isometric Sketches	15	Uses isometric graph correctly; proportions, axes, visible edges, and major features are accurate.	Mostly correct isometric form with minor proportion or feature errors.	Some isometric structure shown but errors affect clarity.	Missing or not isometric.
Multiview Drawings	15	Front/top/right views are correctly aligned, scaled, and show matching geometry.	Views are mostly aligned and matched with minor errors.	Views are incomplete, misaligned, or inconsistent.	Missing or not multiview.
Line Conventions	10	Correctly uses object, hidden, center, construction, dimension, and extension lines where appropriate.	Uses most line types correctly.	Line types are inconsistent or several are missing.	Line conventions not used.
Dimensioning	15	Dimensions are complete, clear, non-duplicated, placed properly, and follow standard practice.	Dimensions are mostly complete with minor placement or clarity issues.	Dimensions are incomplete, crowded, duplicated, or unclear.	Missing dimensions.
Annotations & Design Notes	10	Annotations explain function, material ideas, constraints, and design intent without cluttering the drawing.	Annotations are useful but may be limited.	Annotations are vague or mostly labels only.	Missing.
Aerospace Drawing Package	15	Package is complete, organized, includes required sketches/drawings, and communicates the component clearly.	Package is complete with minor issues.	Package is incomplete or lacks clarity.	Package missing or unacceptable.

Student Self-Check Before Submission

	Requirement	Complete?
☐	Sketching setup and craftsmanship evidence is complete and ready to be graded.	
☐	Concept sketching evidence is complete and ready to be graded.	
☐	Isometric sketches evidence is complete and ready to be graded.	
☐	Multiview drawings evidence is complete and ready to be graded.	
☐	Line conventions evidence is complete and ready to be graded.	
☐	Dimensioning evidence is complete and ready to be graded.	
☐	Annotations and design notes evidence is complete and ready to be graded.	
☐	Aerospace drawing package evidence is complete and ready to be graded.	
☐	All pages/files are named, organized, and submitted correctly.	
☐	Reflection or revision notes are complete.	

Teacher
Use the
evidence
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